

# Palm Oil Retention & Melting System

## Submerged Hydronic Coil Design — Architectural Revision II

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Based on the *VCH Sizing Framework, Second Edition*  
*A Unified Design Methodology for Submerged Hydronic Coil Systems*

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This document presents the second architectural revision of the palm oil storage control system, covering trunk and branch sizing, the need-array priority model, P2 governance, load-responsive operating modes, water-side hydraulics, process duty cycles, aggregate demand budgeting, and a fully specified dynamic water-temperature control law. The work was independently researched, modelled, and authored by Sambridda Ranabhat.

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## Executive Summary

This document presents the second architectural revision of the oil storage control system, superseding the sequential, single-branch design of Revision I. The central change is a **trunk upsize** — from a 1-inch trunk feeding 1-inch branches, to a 1.5 inch trunk capable of supplying **two** 1-inch branches simultaneously at their full rated flow. This single hardware change cascades through the entire control architecture: priority is re-framed as a **two-slot need array** rather than a strict ladder, the P3 (palm oil melting) process gains access to a doubled effective daily energy budget by running two branches in parallel, the PO tank’s governance moves from a hard duty-cycle cap to an operator-trust model backed by a suspicion meter, and — most significantly — the water supply temperature itself becomes a **dynamic, demand-driven quantity** rather than a fixed 70 °C setpoint.

## Headline Results

| Metric                                            | Result              |
|---------------------------------------------------|---------------------|
| Trunk flow advantage over a single branch         | 2×                  |
| T <sub>5</sub> (P1) daily duty period, worst case | 3.67 hr (15.3%)     |
| PO tank (P2) daily duty period                    | 0.59 hr (2.4%)      |
| P3 effective dual-branch energy budget            | ≈43.7 branch-hr/day |
| Palm oil melted per day (P3, dual-delivery)       | ≈4.92 t/day         |
| Combined system throughput (P3 melt + PO tank)    | ≈5.42 t/day         |
| Fulfilment vs. original 2 t/day mandate           | ≈246% (melt only)   |
| Total daily thermal energy demand                 | ≈1081.5 MJ          |
| Average continuous power demand                   | ≈12.5 kW            |

## Key Findings

- **Dual-branch operation roughly doubles P3’s effective daily throughput** relative to Revision I, without any change to coil geometry or water temperature basis — the gain comes entirely from the trunk’s ability to serve two branches at once.
- **Priority is no longer a fixed ladder.** It is now expressed as a *need array*: however many of P1/P2/P3 currently have live demand determines whether the trunk runs a single branch or two branches in parallel. P2 remains capped at a fixed daily allocation; P1 and P3 are unbounded within the array.
- **P2 governance has shifted from hard cutoff to operator trust.** There is no automatic alarm-and-shutoff for exceeding expected PO-tank throughput; instead, an escalating suspicion meter and a hard 2× requested-volume ceiling govern it, with the final stop always requiring a manual action.
- **The system now runs load-responsive, not schedule-driven.** Eco, Efficiency, and Full Throttle modes adjust P3’s operating posture directly off the T<sub>5</sub> level, and the water supply temperature is derived from real-time aggregate demand rather than held at a fixed ceiling.

- **The dynamic water-temperature control law is now fully specified.** A two-point calibration — idle at 40 °C,  $T_5$  worst-case draw at 70 °C — resolves the low-demand inconsistency identified in the working notes and gives the target a clear physical meaning at every operating point.
- **This revision’s worst-case  $T_5$  loss calculation deliberately evaluates the tank at its 55 °C dynamic upper thermal boundary** — a conservative, worst-case basis, not an error.

**On terminology.** An earlier working sketch of this revision used the labels “HSM” (High Supply Mode) and “LSM” (Low Supply Mode) to describe single- vs. dual-branch operation, together with a set of lettered sub-modes. These were exploratory notation only and are **not** carried into the final architecture described here. The underlying concept — the need array determining single- vs. dual-branch operation — is retained and formalised in what follows.

## 1 Trunk and Branch Sizing

### 1.1 Rationale

Revision I’s coil sizing (2-inch bore throughout, 30 m/10 m lengths) left substantial thermal margin relative to P1’s and P2’s actual demand — the water-side film resistance was shown to be a small fraction of overall  $U$ . This revision asks a different question of the same coils: rather than adding more thermal margin, can the *trunk* be upsized so that two branches can be served at once, letting that existing coil margin translate directly into higher system throughput?

### 1.2 Trunk and Branch Cross-Sections

Table 1: Trunk vs. branch sizing.

|                                     | Trunk (1.5-inch)                   | Branch (1-inch)                    |
|-------------------------------------|------------------------------------|------------------------------------|
| Flow area                           | $1.433 \times 10^{-3} \text{ m}^2$ | $6.069 \times 10^{-4} \text{ m}^2$ |
| Design velocity (branch, unchanged) | —                                  | $1.2 \text{ m s}^{-1}$             |

### 1.3 Flow Rate at a Single Branch

$$F_{1\text{-inch}} = A \times V = 6.069 \times 10^{-4} \times 1.2 = 7.283 \times 10^{-4} \text{ m}^3 \text{ s}^{-1} \quad (1)$$

(Consistent with the  $7.315 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$  figure used in the working notes; small variance from rounding of the branch area.)

### 1.4 Trunk Capacity: Powering Two Branches

For the 1.5-inch trunk to supply **two** branches simultaneously, each at the branch design flow rate:

$$F_{\text{trunk,required}} = 2 \times F_{1\text{-inch}} \approx 1.463 \times 10^{-3} \text{ m}^3 \text{ s}^{-1} \quad (2)$$

Solving for the corresponding trunk velocity:

$$V_{\text{trunk}} = \frac{F_{\text{trunk,required}}}{A_{\text{trunk}}} = \frac{1.463 \times 10^{-3}}{1.433 \times 10^{-3}} \approx 1.021 \text{ m s}^{-1} \quad (3)$$

**The 1.5-inch trunk, run at  $\approx 1.021 \text{ m s}^{-1}$ , delivers exactly  $2\times$  the flow of a single 1-inch branch at its normal design velocity.** This is the hardware basis for dual-branch (parallel) operation: the trunk does not need to be oversized further than this to support two simultaneously active branches without starving either one below its individual design flow.

### 1.5 Single-Branch (Reduced Trunk Velocity) Case

When only one branch is active, the trunk need only match that branch's demand:

$$V_{\text{trunk,single}} = \frac{F_{1\text{-inch}}}{A_{\text{trunk}}} = \frac{7.283 \times 10^{-4}}{1.433 \times 10^{-3}} \approx 0.508 \text{ m s}^{-1} \quad (4)$$

(Consistent with the  $\approx 0.511 \text{ m s}^{-1}$  figure in the working notes.) This confirms the trunk is comfortably able to throttle down to serve a single branch without any flow-control conflict.

## 2 Priority Architecture II: The Need Array

### 2.1 From Ladder to Array

Revision I resolved P1/P2/P3 contention through a strict sequential energy budget: P1 and P2 were served first, in full, and P3 received whatever remained of the day. That model assumed only one branch could ever be active at a time. With the trunk now able to serve two branches at once, priority is reframed as a **need array** — the live set of processes currently requesting water:

Table 2: Process-to-tank mapping.

| Slot | Process                                            |
|------|----------------------------------------------------|
| P1   | T <sub>5</sub> trim-heating                        |
| P2   | PO tank (C*) maintenance / dispense                |
| P3   | T <sub>10</sub> palm oil melting (remaining tanks) |
| P4   | Reserved / unused                                  |

- **If the need array contains one active priority**, the trunk runs at single-branch (reduced) velocity, serving that process alone — behaviourally identical to Revision I's single-branch case.
- **If the need array contains two active priorities**, the trunk runs at its full dual-branch velocity (Section 1), serving both simultaneously at their individual full design flow.

- **No hard priority lockout is required between whichever two processes are concurrently active**, since the trunk can serve both at full rate — this is the key structural difference from Revision I, where a second process had to wait even though the physical hardware was not the actual bottleneck.
- P1 remains unbounded in its own demand and is effectively guaranteed service at all times; P2 remains capped (Section 3); P3 has no cap of its own beyond whatever branch-time the array leaves it, but benefits directly from dual-branch operation whenever a second slot is free.

### 3 PO Tank (P2) Governance: Suspicion Meter, Not Alarm

#### 3.1 Rationale for the Change

Revision I capped P2 with a flat arbitration-window limit and treated overrun purely as a scheduling problem. This revision replaces that with an **operator-trust model**: since P2 dispensing is always operator-initiated and operator-confirmed, the system’s role shifts from *preventing* overrun to *flagging* it, escalating only if the operator does not respond.

#### 3.2 The Suspicion Meter

- Expected daily PO-tank throughput remains 500 kg/day (unchanged from Revision I). Sustained draw beyond this threshold raises a **suspicion score**,  $S \in \{0, 1, 2, 3\}$ , which **resets every 24-hour interval**.
- $S=1$ : the HMI queries the operator directly (a confirmation prompt, not yet an alarm).
- $S=2$ : the system raises an audible/visual alarm regarding the suspicion condition.
- $S=3$ : the feed to the PO tank is disabled and an alarm is sounded — the only point at which this pathway forces a shutoff.
- **If there is no HMI response within 5 minutes** of a prompt, the suspicion score is automatically incremented by 1, ensuring the escalation cannot stall indefinitely on operator inaction.

#### 3.3 The Hard Ceiling: $2 \times$ Requested Volume

Independent of the suspicion meter, a second, harder ceiling exists: for any operator-entered dispense quantity  $n_d$  (via HMI), the system permits continued draw up to  $2 \times n_d$  before issuing a warning.

**This ceiling is deliberately manual-shutoff only.** Crossing  $2 \times n_d$  produces a warning, not an automatic cutoff — consistent with the design principle that a live, operator-initiated dispense should never be silently interrupted by the PLC. The suspicion meter is the only pathway in P2’s governance that can autonomously disable the feed, and even that requires either a sustained pattern of excess draw or a prolonged absence of operator response, not a single dispense running long.

## 4 Load-Responsive Operation: Eco, Efficiency, and Full Throttle Modes

### 4.1 Mode Governed by $T_5$ Level

P3's operating posture is tied to the live **level** of  $T_5$  ( $\alpha_S$ ) — the process master (P1) — rather than to  $T_{10}$  tank level or any temperature reading.  $T_5$ 's level is the single clearest signal of how urgently the system as a whole needs melted oil, since every downstream process line draws from  $T_5$  directly.

Table 3:  $T_5$  level-driven operating modes.

| $\alpha_S$ ( $T_5$ level)   | Mode                   | Behaviour                                                               |
|-----------------------------|------------------------|-------------------------------------------------------------------------|
| $\alpha_S \geq 90\%$        | <b>Eco Mode</b>        | Buffer nearly full; very low need. System runs at reduced rate.         |
| $90\% > \alpha_S \geq 50\%$ | <b>Efficiency Mode</b> | Moderate, steady-state need. Standard operating rate.                   |
| $\alpha_S < 50\%$           | <b>Full Throttle</b>   | High need; melt and replenish as aggressively as the need array allows. |

**The 50% Full Throttle threshold is deliberately the same value already used elsewhere in the control architecture** — it is the exact condition ( $\alpha_S < 50\%$ ) that triggers the original batch-release logic ( $T_{10} \rightarrow T_5$  HPP transfer). The moment the system independently decides to release a batch is the same moment P3's operating posture escalates to maximum urgency, so the two mechanisms reinforce each other. The 90% upper threshold is a proposed value and can be tuned during commissioning if a different Eco/Efficiency boundary proves more suitable.

**This mode logic governs P3's operating posture specifically.** P1 and P2 continue to run according to their own demand logic (Sections 6 and 3) regardless of  $T_5$  level.

### 4.2 Mode Selection vs. Dynamic Water Temperature

Mode selection and the demand-driven water temperature of Section 10 answer two different questions and sit in a cause-and-effect relationship, not a comparison:

- **Mode selection is upstream.** It decides *how aggressively P3 should be pulling* — a decision about *production posture*.
- **$E_{\text{total}} \rightarrow T_{\text{water}}$  is downstream.** Whatever P1, P2, and P3 actually demand — P3's demand shaped by whichever mode it is in — gets summed into  $E_{\text{total}}$ , which then decides how hot the water must be to satisfy that combined load. This is a decision about *delivery temperature*.

Concretely: Full Throttle mode increases P3's draw, which increases  $E_\alpha$ , which raises  $E_{\text{total}}$ , which raises  $T_{\text{water}}$ . Mode *feeds into*  $E_{\text{total}}$ ; it does not compete against it. There is no “select whichever is higher” step anywhere in this architecture.



## 5 Water-Side Hydraulics (Retained Basis)

The branch-side hydraulics, coil geometry, and overall  $U$ -values are carried forward unchanged from Revision I — the trunk upsize affects how many branches can run concurrently, not the thermal performance of any individual branch.

Table 4: Retained overall  $U$ -values (2-inch bore, unchanged coil lengths).

| Duty                                                                 | Basis     | $U$ (W/m <sup>2</sup> ·K) |
|----------------------------------------------------------------------|-----------|---------------------------|
| T <sub>5</sub> trim-heat (liquid, $h_o = 138.9$ W/m <sup>2</sup> K)  | 10 m coil | 112.6                     |
| PO tank maintenance (liquid, $h_o = 138.9$ W/m <sup>2</sup> K)       | 30 m coil | 112.5                     |
| T <sub>10</sub> palm oil melt (solid, $h_o = 35$ W/m <sup>2</sup> K) | 30 m coil | 33.0                      |

$$C_w = 2740.7 \text{ W K}^{-1} \quad (\text{per active branch, at } V = 1.2 \text{ m s}^{-1}, \text{ unchanged from Revision I}) \quad (5)$$

$$\varepsilon_{T5} C_w = 205.3 \text{ W K}^{-1} \quad \varepsilon_{PO} C_w = 570.5 \text{ W K}^{-1} \quad \varepsilon_{T10} C_w = 181.5 \text{ W K}^{-1} \quad (6)$$

## 6 P1 — T<sub>5</sub> Duty Cycle: Worst-Case Analysis

### 6.1 Demand Basis: 4 t/day

This revision evaluates T<sub>5</sub> against a **doubled** daily throughput of **4 t/day** (up from 2 t/day in Revision I), reflecting the higher system-level throughput the dual-branch architecture is now sized to support.

$$Q_{T5}(40^\circ\text{C}) = 205.3 \times (70 - 40) = 6160 \text{ W} \quad (\text{max draw}) \quad (7)$$

### 6.2 Sensible Heating Demand

$$E_{\text{sensible}} = 4000 \times 2100 \times 5 = 42.0 \text{ MJ} \quad (8)$$

### 6.3 Worst-Case Standing Loss

**This revision deliberately evaluates T<sub>5</sub>'s standing loss at its 55 °C dynamic upper thermal boundary** (the safety cut-off defined in the original control architecture,  $T_f(x, y, m_p) > 55^\circ\text{C}$ ), rather than at the 42.5 °C operating mid-band used in Revision I. This is a conservative worst-case basis: it asks whether the system still holds up if T<sub>5</sub> is sitting at its hottest permissible boundary.  $\Delta T = 55 - 18.1 \approx 37$  K.

$$E_{\text{loss}} = 0.686 \times 14.25 \times 37 \times 86,400 \approx 32.4 \text{ MJ} \quad (9)$$

$$E_{P1} = 42.0 + 32.4 = 74.4 \text{ MJ/day} \quad (10)$$

## 6.4 Duty Period

The delivery rate is evaluated at the operating mid-band (42.5 °C), consistent with Revision I — only the loss term uses the worst-case boundary temperature:

$$t_{P1} = \frac{74.4 \times 10^6}{205.3 \times 27.5} \approx 13,199 \text{ s} \approx \mathbf{3.67 \text{ hr (15.3\% of 24 hr)}} \quad (11)$$

## 7 P2 — PO Tank Duty Cycle: Unchanged from Revision I

The PO tank's own thermal demand is unaffected by the trunk upsize or the governance changes of Section 3 — those changes govern *how* overrun is handled, not the underlying 500 kg/day, 24–30 °C energy balance itself.

$$E_{P2} = 51.7 \text{ MJ/day} \quad t_{P2} \approx 2107 \text{ s} \approx \mathbf{0.585 \text{ hr (2.4\% of 24 hr)}} \quad (12)$$

## 8 P3 — Dual-Delivery Melting Window

### 8.1 The Doubled Effective Budget

**Modelling approach:** because the trunk can serve two branches at once, P3 is not limited to whatever real-time hours remain after P1 and P2 finish — it can run *two* T<sub>10</sub> coils in parallel during that remaining time. This is modelled as an effective **48 branch-hour** daily pool (2 branches × 24 hr), from which P1's and P2's own single-branch consumption is subtracted. The remainder is P3's effective budget, realisable either as two tanks melting in parallel for half the real time, or one tank melting for the full equivalent duration — the total energy delivered is the same either way.

$$t_{P3, \text{budget}} = 48 - (t_{P1} + t_{P2}) = 48 - (3.67 + 0.585) \approx \mathbf{43.7 \text{ branch-hr/day}} \quad (13)$$

### 8.2 Melt Rate (Unchanged Coil Basis)

$$Q_{T10}(36^\circ\text{C}) = 181.5 \times (70 - 36) \approx 6172 \text{ W} \quad (14)$$

Same  $NTU_{T10} = 0.0685$ ,  $\varepsilon_{T10} = 0.0662$  as Revision I — coil geometry and water temperature basis for this duty are unchanged.

### 8.3 Mass Melted

$$E_{P3} = 6172 \text{ W} \times (43 \times 3600) \text{ s} \approx 955.4 \text{ MJ} \quad (15)$$

$$m_{P3} = \frac{955.4 \times 10^6}{194,200} \approx \mathbf{4919.8 \text{ kg/day (4.92 t/day)}} \quad (16)$$

**Cross-check:** melting two full 10 kL tanks in parallel ( $m = 17\,800 \text{ kg}$ ) at twice the coil power ( $2 \times 6172 \text{ W}$ ) takes the *same* 77.8 hr as melting one tank alone at single-coil power — mass and power both double, so melt time is unchanged. This confirms the dual-branch gain is a genuine doubling of throughput, not merely a faster single-tank cycle.

## 9 Aggregate Daily Demand and System Output

### 9.1 Total Daily Thermal Energy Demand

$$E_{\text{total}} = E_{P1} + E_{P2} + E_{P3} = 74.4 + 51.7 + 955.4 \approx 1081.5 \text{ MJ/day} \quad (17)$$

$$P_{\text{avg}} = \frac{1081.5 \times 10^6}{86,400} \approx 12.5 \text{ kW continuous average} \quad (18)$$

### 9.2 System Throughput and Fulfilment

Table 5: Combined daily throughput.

| Stream                              | Daily mass                                                 |
|-------------------------------------|------------------------------------------------------------|
| Palm oil melted (P3, dual-delivery) | 4919.8 kg                                                  |
| Palm olein dispensed (P2)           | 500.0 kg                                                   |
| <b>Combined system throughput</b>   | <b>5419.8 kg (<math>\approx 5.42 \text{ t/day}</math>)</b> |

$$\text{Fulfilment (melt only)} = \frac{4919.8}{2000} \times 100\% \approx \mathbf{246\%} \quad (19)$$

$$\text{Fulfilment (combined)} = \frac{5419.8}{2000} \times 100\% \approx \mathbf{271\%} \quad (20)$$

**Against the original 2 t/day mandate, this revision delivers roughly 2.5× to 2.7× the required throughput,** a direct consequence of the trunk’s dual-branch capability rather than any change to individual coil sizing.

**Note on the  $\approx 6$  t/day estimate.** The rigorous figure worked through here is  $\approx 5.42$  t/day, slightly below the  $\approx 6$  t/day ballpark from early scoping. This gap is not an error — it reflects a **deliberately conservative modelling choice**: P3’s 48 branch-hour pool is reduced by the full raw duration of P1’s and P2’s duty periods ( $t_{P1} + t_{P2} = 4.25$  branch-hr), regardless of how much overlap actually occurs day to day. A model that credits P3 for partial overlap (P3 usually still has one branch free while P1 or P2 draws its other slot) would recover most of the remaining gap. The figure adopted here is the conservative floor by design, not the best-case ceiling.

## 10 Dynamic Water Temperature Control

### 10.1 Motivation

Both prior revisions held the water supply at a fixed  $70^\circ\text{C}$ . This is now identified as unnecessary: the water only needs to be as hot as the *current* aggregate demand requires. This section formalises a control law that derives the water target from real-time load rather than a fixed setpoint, then gates the solar loop and auxiliary driver against that dynamic target.

### 10.2 Aggregate Demand Function

The total instantaneous energy demand is the sum of whichever processes are currently active:

$$E_{\text{total}} = E_\alpha + E_5 + E_{C^*} \quad (21)$$

#### $E_\alpha$ — Generic $T_{10}$ Melting-Tank Demand

For a tank at current temperature  $t_\alpha$  and mass  $m_\alpha$ , using  $C_{p,s} = 2000$ ,  $L_f = 150,000$ ,  $C_{p,l} = 2100$  J/kg.K, melt point  $36^\circ\text{C}$ , delivery target  $40^\circ\text{C}$ :

$$E_\alpha(t_\alpha, m_\alpha) = \begin{cases} [2000(36 - t_\alpha) + 150,000 + 2100 \times 4] m_\alpha, & t_\alpha \leq 36^\circ\text{C} \\ 2100 (40 - t_\alpha) m_\alpha, & 36^\circ\text{C} < t_\alpha \leq 40^\circ\text{C} \\ 0, & t_\alpha > 40^\circ\text{C} \end{cases} \quad (22)$$

**The step at exactly  $36^\circ\text{C}$  is a genuine physical discontinuity, not a modelling error.** Just below the melt point, the tank still requires the full 150 000 J/kg of latent heat (remaining demand  $\approx 158\,400$  J/kg); once that heat has been absorbed, remaining demand drops sharply to liquid-phase sensible heating only ( $\approx 8400$  J/kg). Standing heat loss is deliberately **excluded** from  $E_\alpha$  — it is already accounted for as a separate daily energy term in Sections 8 and 9, and including it here would double-count it.

#### $E_5$ — $T_5$ Trim-Heating Demand

$$E_5(T_F, T_5, M_5, T) = 2100 M_5 (T_F - T_5) + 0.686 \times 14.25 (T_5 - T) \quad (23)$$

where  $T_F$  is the incoming oil feed temperature and  $T$  is ambient. Unlike  $E_\alpha$ ,  $E_5$  carries its own standing-loss term explicitly ( $0.686 \times 14.25 = 9.78 \text{ W K}^{-1}$ ,  $T_5$ 's insulated jacket coefficient  $\times$  area from Section 6), since  $T_5$ 's loss was not separately budgeted elsewhere in this instantaneous-demand context the way  $T_{10}$ 's was.

### $E_{C^*}$ — PO Tank Demand

$$E_{C^*}(T_c, M_c, T) = 2100 M_c (30 - T_c) + 59.08 (T_c - T) \quad (24)$$

The loss coefficient  $59.08 \text{ W K}^{-1}$  is  $U_{\text{cyl}} A_{\text{cyl}} + U_{\text{base}} A_{\text{base}}$  from the PO tank loss calculation. This term is essential: the PO tank's standing loss outweighs its sensible duty by roughly 7:1 (tank is uninsulated), and omitting it would badly understate instantaneous demand.

**On the linearity assumption in the loss coefficients.** Both  $9.78 \text{ W K}^{-1}$  ( $T_5$ ) and  $59.08 \text{ W K}^{-1}$  (PO tank) were derived from natural-convection correlations evaluated at one specific  $\Delta T$ . Physically,  $h$  scales roughly as  $\Delta T^{1/4}$  in the laminar regime, so writing loss as a simple linear function of  $(T_{\text{tank}} - T)$  is an approximation, not an exact relation. It was kept deliberately: (1) it allows real-time evaluation in a 10 s-cycle PLC without re-solving a Nusselt correlation each cycle, and (2) the error is small over the narrow operating bands each tank spans —  $T_5$ :  $40\text{--}45^\circ\text{C}$ ; PO tank:  $24\text{--}30^\circ\text{C}$ . Insulating the PO tank (recommended in Section 3) would substantially reduce the magnitude of the loss term and make this approximation safer regardless.

### Assembled Total

$$E_{\text{total}}(t_\alpha, m_\alpha, T_F, T_5, M_5, T_c, M_c, T) = E_\alpha + E_5 + E_{C^*} \quad (25)$$

## 10.3 From Demand to Water Temperature

Demand  $E_{\text{total}}$  is evaluated on a 10 s refresh window and converted to an instantaneous power  $P$ . The water target temperature is then proportional to  $P$ , anchored at **two known endpoints** — idle and  $T_5$ 's worst-case draw:

$$T_{\text{water}} = 40 + \frac{70 - 40}{6.16} P = 40 + 4.87 P \text{ [kW]}, \quad \text{clamped to } [40^\circ\text{C}, 70^\circ\text{C}] \quad (26)$$

where 6.16 kW is  $T_5$ 's own established worst-case draw ( $Q_{T_5}(40^\circ\text{C})$ , Section 6). Anchoring at both endpoints — idle ( $P = 0 \rightarrow 40^\circ\text{C}$ ) and  $T_5$  worst case ( $P = 6.16 \text{ kW} \rightarrow 70^\circ\text{C}$ ) — resolves the low-demand inconsistency of an earlier proportional law through the origin.

At  $P = 1 \text{ kW}$ :  $T_{\text{water}} = 44.9^\circ\text{C}$  — a sensible result just above the floor. At  $P \geq 6.16 \text{ kW}$ : the target saturates at  $70^\circ\text{C}$ , meaning the system reaches full design temperature any time aggregate demand is comparable to a single  $T_5$  worst-case event.

## 10.4 Solar and Auxiliary Gating

With the dynamic target  $T_w$  established, the solar loop and auxiliary driver are gated purely on whether each can meet it:

- $T_{\text{solar}} > T_w$ : run the through-solar cycle (solar alone meets current demand).

- $T_{\text{solar}} < T_w$ : bypass solar.
- The auxiliary driver activates **only** when the system’s computed demand requires it — not on a fixed schedule. Whenever the auxiliary is active,  $T_w > T_{\text{solar}}$  holds by definition, so solar bypass is always concurrent with auxiliary operation.

## 10.5 Exchanger Outlet Water Temperature

The water returning from each exchanger retains most of its heat, since the low per-pass effectiveness of the batch-heating model means only a small fraction of the available driving temperature is extracted per loop:

$$T_{\text{water,out}} = T_{\text{hot}} - \varepsilon (T_{\text{hot}} - T_{\text{tank}}) \quad (27)$$

evaluated at  $T_{\text{hot}} = 70^\circ\text{C}$ , using  $\varepsilon_{T5} = 0.0749$  and  $\varepsilon_{T10} = 0.0662$ .

Table 6:  $T_5$  outlet water temperature — worst and best case.

| Condition                             | $T_{\text{tank}}$  | $T_{\text{water,out}}$ |
|---------------------------------------|--------------------|------------------------|
| Worst (just triggered, P1 fires)      | $40^\circ\text{C}$ | $67.75^\circ\text{C}$  |
| Best (near target, P1 about to clear) | $44^\circ\text{C}$ | $68.05^\circ\text{C}$  |

Table 7:  $T_{10}$  outlet water temperature — worst, ambient, and best case fill.

| Condition                          | $T_{\text{tank}}$  | $T_{\text{water,out}}$ |
|------------------------------------|--------------------|------------------------|
| Worst (extreme cold winter fill)   | $2^\circ\text{C}$  | $65.50^\circ\text{C}$  |
| Ambient (Kathmandu annual average) | $18^\circ\text{C}$ | $66.56^\circ\text{C}$  |
| Best (mild fill condition)         | $24^\circ\text{C}$ | $66.96^\circ\text{C}$  |

### All five outlet temperatures sit within roughly 2–4.5 K of the $70^\circ\text{C}$ supply.

This is the expected consequence of the deliberately low per-pass effectiveness — only a small fraction of the available driving temperature is extracted per coil pass, by design. The return line consequently carries most of its heat back to the thermal battery rather than losing it to the tank, which is consistent with and confirms the batch-heating assumption used throughout this document.

**Interpretation of the  $T_{10}$  reference points.**  $18^\circ\text{C}$  matches the Kathmandu weighted annual ambient average used throughout this document.  $2^\circ\text{C}$  is an extreme cold-season worst-case fill, and  $24^\circ\text{C}$  a mild best-case fill. These are bounding reference points for this outlet-temperature sweep only — confirm this reading before treating the 2/24 $^\circ\text{C}$  figures as design inputs elsewhere.

## 11 Conclusions and Recommended Next Steps

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1. **The trunk upsize is the single highest-leverage change in this revision.** Without altering any coil, tank, or water temperature specification, enabling two branches in parallel roughly doubles P3’s effective daily throughput, lifting combined system output to  $\approx 5.42$  t/day ( $\approx 271\%$  of the original 2 t/day mandate).
2. **The need-array priority model is a genuine simplification over a strict ladder,** since it lets the hardware’s actual capacity — two branches — govern how much of the system runs concurrently, rather than an artificial single-branch assumption.
3. **P2’s shift to an operator-trust governance model is appropriate given its use case:** a live, technician-supervised dispense should not be silently interrupted, and the suspicion meter plus  $2 \times n_d$  ceiling together provide layered but non-disruptive oversight.
4. **The dynamic water-temperature control law is now fully specified.** Two-point calibration at idle and  $T_5$ ’s worst-case draw resolves the low-demand inconsistency identified in development, and gives the water target a clear physical meaning at every operating point.
5. **Mode selection and demand-driven water temperature are not competing signals.** Mode (governed by  $T_5$  level) decides how aggressively P3 draws; that shapes  $E_\alpha$ , which feeds into  $E_{\text{total}}$ , which sets  $T_{\text{water}}$ . The relationship is causal, not comparative — there is no “select the higher value” step anywhere in this architecture.
6. **The worst-case basis used throughout should be labelled consistently.**  $T_5$ ’s loss at  $55^\circ\text{C}$  and P3’s flat floor subtraction are both conservative-by-design choices; day-to-day operational figures will typically run better than either, and the distinction should be explicit wherever these calculations are reused.
7. **Remaining open items:** (a) confirm the  $2/24^\circ\text{C}$   $T_{10}$  outlet reference points against their intended design meaning; (b) consider insulating the PO tank (standing recommendation from Section 3), which would improve efficiency and reduce  $E_{C^*}$ ’s sensitivity to operating temperature; (c) if a less conservative P3 throughput estimate is desired, revisit the overlap-based refinement (crediting P3 for the fact that P1 and P2 rarely draw at the exact same instant) as an alternative to the current flat floor-subtraction.

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